

Jianke Yu 俞鉴珂, PhD

PhD (2026) in Graph Machine Learning & Databases

Hangzhou, China

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RESEARCH AREAS

Machine learning for graph-structured data and database systems: graph neural network architectures for incomplete, heterogeneous and temporal graphs (fraud detection, malware detection, graph similarity computation), and learning-based methods that accelerate database and data-mining algorithms.

Keywords: Graph Neural Networks, Graph Databases, AI for Databases, Fraud Detection, Knowledge Graphs

EDUCATION

PhD, Information Technology · University of Technology Sydney 2023.07 – 2026.09

Conferred 9 September 2026. Supervised by Prof. Ying Zhang, Prof. Lu Qin and Dr Hanchen Wang. Thesis: "Fraud detection on the customer-product bipartite graph" (learning-based algorithms for graph-structured data and database systems).

M.Eng, Electronic Information · Zhejiang Gongshang University 2020.09 – 2023.06

Supervised by Dr Xiaoyang Wang. Research on graph representation learning, fraud detection and graph similarity computation.

B.Eng, Electronic Information Engineering · Zhejiang Gongshang University 2016.09 – 2020.06

APPOINTMENTS & EXPERIENCE

Research Assistant · University of Technology Sydney 2026.07 – Present

Casual research assistant on the ARC Discovery Project led by Prof. Ying Zhang. Graph neural networks, graph similarity search and learning-based database algorithms. Ongoing.

Research Intern · Token Foundry, Alibaba Token Hub, Alibaba Group 2025.12 – Present

Research on graph learning methods for industrial data systems.

Guest Lecturer · University of Technology Sydney 2025.08

Invited guest lecturer for UTS 43023 Emerging Topics in AI, delivering the Large-Scale Graph topic block and guiding students through the implementation of graph models (confirmed by a UTS letter of confirmation, 27 Aug 2025).

Research Intern · DAMO Academy / Alibaba Cloud, Alibaba Group 2022.07 – 2023.04

Fraud detection on e-commerce platforms (work later published in KDD 2023 and IEEE TKDE).

Research Intern · Taobao, Alibaba Group 2021.11 – 2022.03

Graph-based user modelling and risk analysis.

SELECTED PUBLICATIONS (FIRST-AUTHOR CCF-A)

Overall: 19 peer-reviewed papers (10 SCI journal papers, 5 in JCR Q1; 9 conference papers) — 7 CCF-A, 6 CCF-B, 4 CCF-C; 6 as first author; 228 citations (Google Scholar).

[1] MGDN: A Graph of Graphs Neural Network for Malware Detection. *IEEE Transactions on Knowledge and Data Engineering*, 2026. DOI: 10.1109/TKDE.2026.3689176 **1st author** CCF-A CAS Zone 1 Top JCR Q1

[2] SARL: A Scalable Attribute-Informed Representation Learning Framework for Bipartite Graphs. *IEEE Transactions on Knowledge and Data Engineering*, 2026. DOI: 10.1109/TKDE.2026.3735058 **1st author** CCF-A CAS Zone 1 Top JCR Q1

[3] Temporal Insights for Group-Based Fraud Detection on e-Commerce Platforms. *IEEE Transactions on Knowledge and Data Engineering*, 2024. DOI: 10.1109/TKDE.2024.3485127 **1st author** CCF-A CAS Zone 1 Top JCR Q1

[4] Group-based Fraud Detection Network on e-Commerce Platforms. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023. DOI: 10.1145/3580305.3599836 **1st author** CCF-A

SCHOLARSHIPS

National Award for Outstanding Self-Financed Students Abroad (2025–2026) · China Scholarship Council 2026

National-level award for outstanding self-financed students abroad.

UTS ARC Discovery Project Scholarship (Full Scholarship) · University of Technology Sydney / Australian Research Council 2023

UTS International Research Scholarship · University of Technology Sydney 2023

HONORS & AWARDS

UTS FEIT HDR Excellence Award — Individual Category, Distinction · University of Technology Sydney, Faculty of Engineering and IT 2025

Faculty-level award (Distinction) for research excellence during HDR candidature.

Excellent Practical Achievement Award for Professional Master Students · Education Department of Zhejiang Province 2023

Provincial award (2022 round) for outstanding professional master's practice achievements.

Second Prize, 4th Zhejiang Province University Student Robotics Competition · Zhejiang Province 2019

Second Prize, "Internet+" Innovation and Entrepreneurship Competition · Zhejiang Gongshang University 2019

ACADEMIC ACTIVITIES & SERVICE

Scalable Big Data Computation · Seminar, School of Data Science, The Chinese University of Hong Kong, Shenzhen 18 Apr 2026

Journal Peer Reviewer · IEEE Transactions on Knowledge and Data Engineering (TKDE); Computers & Security; Information Processing & Management 2025–

Conference Peer Reviewer · ADC, APWeb-WAIM and other international conferences

Platform Administrator · Research Computing Platform, Prof. Ying Zhang's Team, Zhejiang Gongshang University 2021–

Responsible for the group's research computing platform and graph data infrastructure: deployment and scheduling of GPU clusters and multi-GPU LLM inference services, research accounts and compute resource management, monitoring and alerting; development and maintenance of a self-built graph data storage and exploration base and a self-built cluster management platform (both closed-source, in internal use).

TEACHING

Guest Lecturer · 43023 Emerging Topics in AI, University of Technology Sydney Aug 2025

Invited guest lecturer for the Large-Scale Graph topic block; confirmed by a UTS letter of confirmation (subject coordinator Dr Yi Zhang, 27 Aug 2025).

Teaching Assistant · COMP9311 Database Systems, UNSW Sydney Term 3 2023

Teaching Assistant · DATA1001 Introduction to Data Science, UNSW Sydney

Term 2 2024, Term 2 2025